

TABLE OF CONTENTS
S-404 ANCHOR BOLT WORKS

	Page No.
1. SCOPE OF WORK.....	S.404-2
2. CODES AND STANDARDS	S.404-2
3. MATERIAL: ANCHOR BOLTS, NUTS AND WASHERS	S.404-3
4. ANCHOR BOLT SLEEVES.....	S.404-3
5. SETTING	S.404-3
6. COATING AND PROTECTION.....	S.404-4
7. QUALITY ASSURANCE/QUALITY CONTROL	S.404-4

S.404 ANCHOR BOLT WORKS

1. SCOPE OF WORK

- 1.1 The scope of anchor bolt shall include anchor bolts required for structural steel, gallery steel, auxiliary steel, equipment bases etc. as applicable to Bidder's work as indicated in Part 4, subsection 1.2 "Summary of Work" and S-001 "General Design Requirement of Structure, Civil and Building Works".
- 1.2 The Bidder shall design, manufactures, test and supply and install all anchor bolts included in the Contract.
- 1.3 All anchor bolts including templates and bolt setting diagrams shall be shipped to the Site early to meet the schedule of anchor bolts setting in concrete foundation.
- 1.4 Anchor bolts, nuts, washers and anchor bolt sleeves shall conform to the applicable requirements of this specification, and the Standard drawings.

2. CODES AND STANDARDS

- 2.1 The Bidder shall comply with the requirements of Part 4, Subsection 2.5 "Codes and Standards".
- 2.2 The codes and standards of the following organizations extracted from the Part 4, Subsection 2.5 "Codes and Standards" are specifically applicable to the design, manufacturer and testing of the work included in this Specifications:

ANSI	American National Standards Institute
ASTM	American for Testing and Materials
JIS	Japanese Industrial Standards

3. MATERIAL: ANCHOR BOLTS, NUTS AND WASHERS

- 3.1 Nominal: ASTM A36 for threaded rods and ASTM A307 for headed bolts and for all nuts.
- 3.2 High Strength: ASTM A193, Grade B7 for threaded rods and headed bolts, with ASTM A194, Grade 2 ANSI Series nuts and ASTM A325 for washers. Hardened washers shall be provided with high strength anchor bolts.
- 3.3 Bolts and nuts shall be of the coarse-thread series. Bolts shall have 2A tolerances and nuts shall have 2B tolerances. Bolts shall conform to ANSI B1.1. Nuts shall be heavy semi finished hexagon nuts conforming to ANSI B18.2.2.
- 3.4 Nuts and washers for galvanized structures shall be galvanized in conformance with ASTM Specification A153.

3.5 Report and Certification

Mill Test Report: Two copies of the mill test report, as required applicable Codes or Standards, shall be submitted by the Bidder to the Owner.

4. ANCHOR BOLT SLEEVES

- 4.1 Anchor bolt sleeves shall be made of standard weight steel pipe conforming to ASTM A120 or made of polyethylene plastic.
- 4.2 Plastic Plugs: For anchor bolts with pipe sleeves, provide plastic plugs at top of sleeves. Provide plugs with center opening 3 mm smaller in diameter than bolt size to ensure weather tight fit. Openings shall be shop or field punched or drilled. Plugs shall be maintained until the foundation is prepared for grouting prior to setting base plates and/or equipment.

5. SETTING

- 5.1 Templates shall be provided for all anchor bolts. Templates shall be carefully designed and constructed of sufficiently thick and rigid material so as to hold the bolts to line and elevation as set.
- 5.2 Anchor bolts shall be accurately located and set to line and elevation before concrete is poured and the accuracy of position of anchor bolts shall be

maintained throughout the placing and finishing of concrete by the erector of the anchor bolts.

5.3 Column anchor bolts will be erected to the following tolerance:

5.3.1 Individual anchor bolts: 3 mm

5.3.2 Anchor bolts group: 6 mm

5.4 Heating and welding: Heating of any anchor bolt in the field for bending or other purposes will not be permitted, except that nuts for threaded rods (A36) or headed bolts (A307) may be tack welded to the plate washers used for embedment in concrete.

5.5 Anchor bolt threads shall be protected against damage. Damaged anchor bolts shall be repaired or replaced to the satisfaction of the Owner.

5.6 Grouting in anchor sleeves shall be installed using non-shrink grout only. Concrete mixes (regardless of any additives) in lieu of non-shrink grout are unacceptable.

6. COATING AND PROTECTION

All anchor bolts shall be protected with antirust compound suitable for the environmental conditions prior to shipment.

7. QUALITY ASSURANCE/QUALITY CONTROL

7.1 Quality assurance/assurance control shall conform shall conform to the requirements of Part 4

7.2 The quality control procedures and documentation shall include but not be limited to the following:

7.2.1 Welding

7.2.2 Mill certificate of material

7.2.3 Non-destructive testing.

7.2.4 Packaging and shipping.

7.2.5 Shop test procedures.

7.2.6 Cleaning and painting.

(Last page of S-404)